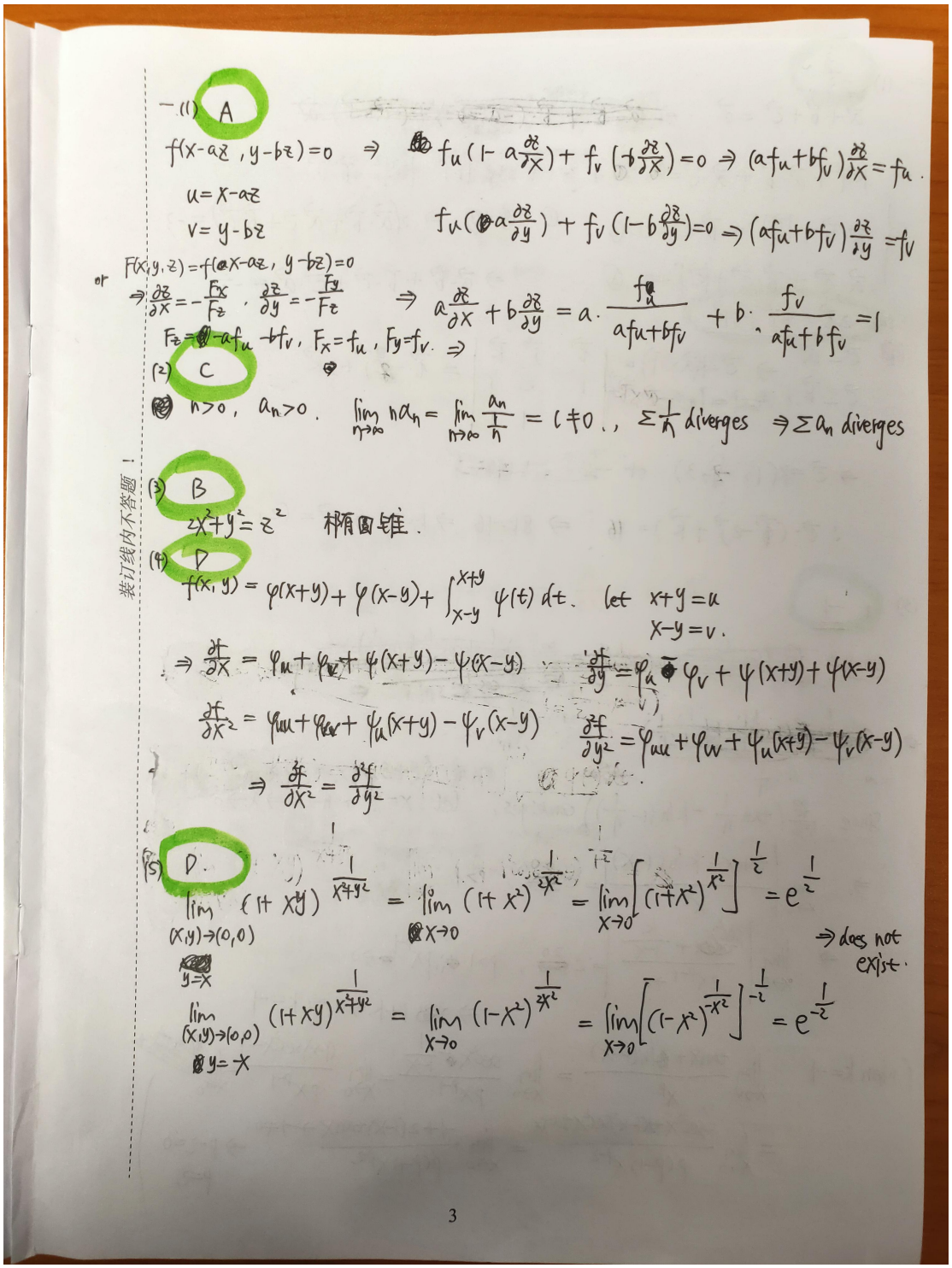
答案



一.(1) A

$f(x-az, y-bz)=0 \Rightarrow f_u(1-a\frac{\partial z}{\partial x})+f_v(-b\frac{\partial z}{\partial x})=0 \Rightarrow (af_u+bf_v)\frac{\partial z}{\partial x}=f_u$

$u=x-az$

$v=y-bz$

$f_u(-a\frac{\partial z}{\partial y})+f_v(1-b\frac{\partial z}{\partial y})=0 \Rightarrow (af_u+bf_v)\frac{\partial z}{\partial y}=f_v$

or $F(x,y,z)=f(x-az, y-bz)=0$

$\Rightarrow \frac{\partial z}{\partial x}=-\frac{F_x}{F_z}, \frac{\partial z}{\partial y}=-\frac{F_y}{F_z} \Rightarrow a\frac{\partial z}{\partial x}+b\frac{\partial z}{\partial y}=a\cdot\frac{f_u}{af_u+bf_v}+b\cdot\frac{f_v}{af_u+bf_v}=1$

$F_z=-af_u-bf_v, F_x=f_u, F_y=f_v. \Rightarrow$

(2) C

$n>0, a_n>0. \lim_{n\to\infty} na_n=\lim_{n\to\infty}\frac{a_n}{\frac{1}{n}}=1\neq 0., \sum\frac{1}{n}$ diverges $\Rightarrow \sum a_n$ diverges

(3) B

$z^2+y^2=z^2$ 椭圆锥.

(4) D

$f(x,y)=\varphi(x+y)+\varphi(x-y)+\int_{x-y}^{x+y}\psi(t)\,dt.$ let $x+y=u$, $x-y=v.$

$\Rightarrow \frac{\partial f}{\partial x}=\varphi_u+\varphi_v+\psi(x+y)-\psi(x-y)$ $\frac{\partial f}{\partial y}=\varphi_u-\varphi_v+\psi(x+y)+\psi(x-y)$

$\frac{\partial f}{\partial x^2}=\varphi_{uu}+\varphi_{uv}+\psi_u(x+y)-\psi_v(x-y)$ $\frac{\partial f}{\partial y^2}=\varphi_{uu}+\varphi_{uv}+\psi_u(x+y)-\psi_v(x-y)$

$\Rightarrow \frac{\partial f}{\partial x^2}=\frac{\partial f}{\partial y^2}$

(5) D.

$\lim_{(x,y)\to(0,0)}(1+xy)^{\frac{1}{x^2+y^2}}=\lim_{x\to 0}(1+x^2)^{\frac{1}{2x^2}}=\lim_{x\to 0}\left[(1+x^2)^{\frac{1}{x^2}}\right]^{\frac{1}{2}}=e^{\frac{1}{2}}$

$\Rightarrow$ does not exist.

$y=x$

$\lim_{(x,y)\to(0,0)}(1+xy)^{\frac{1}{x^2+y^2}}=\lim_{x\to 0}(1-x^2)^{\frac{1}{2x^2}}=\lim_{x\to 0}\left[(1-x^2)^{\frac{1}{-x^2}}\right]^{-\frac{1}{2}}=e^{-\frac{1}{2}}$

$y=-x$

装订线内不答题！

(1) 二. (1) $-\frac{3}{2}$

$\vec{a}+\vec{b}+\vec{c}=\vec{0}$

$\Rightarrow |\vec{a}|^2+\vec{a}\cdot\vec{b}+\vec{a}\cdot\vec{c}=0$ ①

$\begin{cases} \vec{a}\cdot\vec{b}+|\vec{b}|^2+\vec{b}\cdot\vec{c}=0 & ② \\ \vec{a}\cdot\vec{c}+\vec{b}\cdot\vec{c}+|\vec{c}|^2=0 & ③ \end{cases}$

① + ② + ③ $\Rightarrow 2(\vec{a}\cdot\vec{b}+\vec{a}\cdot\vec{c}+\vec{b}\cdot\vec{c})=-3$

$\Rightarrow \vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}=-\frac{3}{2}$

(2) $2\vec{i}-4\vec{j}+6\vec{k}$

$\vec{c}\perp\vec{a}$ $\Rightarrow \vec{c}=k(\vec{a}\times\vec{b})=k\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 1 \\ -1 & 1 & 1 \end{vmatrix}=k(\vec{i}-2\vec{j}+3\vec{k})$

$\vec{c}\perp\vec{b}$

$\Rightarrow \vec{c}=k(1,-2,3)$

$\because \vec{c}\cdot(\vec{i}-2\vec{j}+\vec{k})=16 \Rightarrow 8k=16 \Rightarrow k=2 \Rightarrow \vec{c}=(2,-4,6)$

(3) -1

Since $\sum_{n=2}^{\infty}\left(\tan\frac{1}{n}-k\ln\left(1-\frac{1}{n}\right)\right)$ converges, let $x=\frac{1}{n}$, $n\to\infty \Rightarrow x\to 0$

$\Rightarrow \lim_{x\to 0}\left|\frac{\tan x-k\ln(1-x)}{x^p}\right|=c\geq 0$, $p>1$

$\Rightarrow \lim_{x\to 0}\left|\frac{\sec^2 x+\frac{k}{1-x}}{px^{p-1}}\right|=c\geq 0$, $p>1$ 且 $px^{p-1}\to 0$

$\Rightarrow 1+k\to 0 \Rightarrow k=-1$

(when $k=-1$ $\lim_{x\to 0}\frac{\tan x+\ln(1-x)}{x^p}=\lim_{x\to 0}\frac{\sec^2 x-\frac{1}{1-x}}{px^{p-1}}=\lim_{x\to 0}\frac{(1-x)\sec^2 x-1}{px^{p-1}}$

$=\lim_{x\to 0}\frac{-\sec^2 x+2(1-x)\sec^2 x\tan x}{p(p-1)x^{p-2}}=\lim_{x\to 0}\frac{-1+2(1-x)\tan x}{p(p-1)x^{p-2}}\to -1\neq 0$

$\Rightarrow p-2=0$

$p=2$

裝訂線內不答題

(4) 1

$y = f(x) = \sin x \Rightarrow f'(x) = \cos x,\ f''(x) = -\sin x$

$$K = \frac{|f''(x)|}{(1+(f'(x))^2)^{\frac{3}{2}}} = \frac{|-\sin x|}{(1+\cos^2 x)^{\frac{3}{2}}} = \frac{|\sin x|}{(2-\sin^2 x)^{\frac{3}{2}}}$$

when $|\sin x|$ max, $\sin^2 x$ is max

$\Rightarrow 2-\sin^2 x$ is min.

$\Rightarrow (2-\sin^2 x)^{\frac{3}{2}}$ is min.

$\Rightarrow \frac{1}{(2-\sin^2 x)^{\frac{3}{2}}}$ is max

$\Rightarrow |\sin x| \cdot \frac{1}{(2-\sin^2 x)^{\frac{3}{2}}}$ is max

$\Rightarrow K$ is max

$\Rightarrow |\sin x| = 1 \Rightarrow K = 1$

(5) $2-2\ln 2$

$(z+y)^x = xy \Rightarrow x\ln(z+y) = \ln xy$

$\Rightarrow \ln(z+y) + x \cdot \frac{1}{z+y} \cdot \frac{\partial z}{\partial x} = \frac{1}{xy} \cdot y$

let $x=1,\ y=2,\ z=0$

$\Rightarrow \ln 2 + \frac{1}{2} \cdot \frac{\partial z}{\partial x} = 1 \Rightarrow \frac{\partial z}{\partial x} = 2-2\ln 2$

or let $F(x,y,z) = x\ln(z+y) - \ln xy$

$\Rightarrow F_x = \frac{\ln(z+y)}{} - \frac{y}{xy} = \ln 2 - 1$

$F_z = \frac{x}{z+y} = \frac{1}{2}$

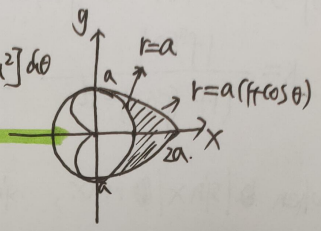
$\Rightarrow \frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = 2-2\ln 2.$

装订线内不答题！

3.

(1)

$$A=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{1}{2}\left[a^2(1+\cos\theta)^2-a^2\right]d\theta \quad \text{or} \quad A=2\int_0^{\frac{\pi}{2}}\frac{1}{2}\left[a^2(1+\cos\theta)^2-a^2\right]d\theta$$



$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{1}{2}\left[2a^2\cos\theta+a^2\cos^2\theta\right]d\theta$$

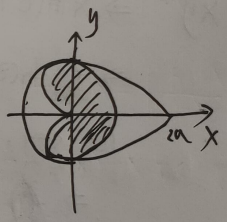
$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{1}{2}\left[2a^2\cos\theta+a^2\cdot\frac{1+\cos 2\theta}{2}\right]d\theta.$$

$$=\frac{1}{2}\left(2a^2\sin\theta+\frac{1}{2}a^2\theta+\frac{a^2}{4}\sin 2\theta\right)\Big]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$=2a^2+\frac{a^2\pi}{4}=\left(2+\frac{\pi}{4}\right)a^2$$

(2)

$$A_1=2\int_0^{\pi}\frac{1}{2}\left[a^2(1+\cos\theta)^2\right]$$



$$=\int_0^{\pi}\left(a^2+2a^2\cos\theta+a^2\cos^2\theta\right)d\theta.$$

~~$a^2\theta+2a^2\sin\theta$~~

$$=\int_0^{\pi}\left(a^2+2a^2\cos\theta+a^2\cdot\frac{1+\cos 2\theta}{2}\right)d\theta.$$

$$=a^2\theta+2a^2\sin\theta+\frac{a^2}{2}\theta+\frac{a^2}{4}\sin 2\theta\Big]_0^{\pi}$$

$$=\frac{3}{2}a^2\pi$$

$$\Rightarrow A_2=A_1-A=\frac{3}{2}a^2\pi-\left(2+\frac{\pi}{4}\right)a^2=\frac{5}{4}\pi a^2-2a^2=\left(\frac{5}{4}\pi-2\right)a^2.$$

装订线内不答题！

4. $\vec{r}(t) = f(t)\vec{i} + g(t)\vec{j} + h(t)\vec{k}$. , $f(t) = \int_0^t \cos(x^2)\,dx$, $g(t) = -t\cos t$

$$h(t) = \sum_{n=1}^{\infty} \frac{t^n}{n}$$

$$\Rightarrow h'(t) = \sum_{n=1}^{\infty} t^{n-1} = \sum_{n=0}^{\infty} t^n = \frac{1}{1-t}$$

$f'(t) = \cos(t^2)$

$g'(t) = -\cos t + t\sin t$

$\Rightarrow f'(0) = 1$, $g'(0) = -1$, $h'(0) = 1$

$\Rightarrow r'(0) = (1, -1, 1)$

5. (1) $f(x,y) = \begin{cases} y\sin\frac{1}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} y\sin\frac{1}{x^2+y^2} = 0 \quad \text{since } \left|\sin\frac{1}{x^2+y^2}\right| \le 1$$

$0 \le |f(x,y)| \le |y| \to 0$.

(2) $f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{0-0}{h} = 0$

$$f_y(0,0) = \lim_{h\to 0} \frac{f(0,h) - f(0,0)}{h} = \lim_{h\to 0} \frac{h\sin\frac{1}{h^2} - 0}{h} = \lim_{h\to 0} \sin\frac{1}{h^2} \text{ does not exist.}$$

6. (1) $\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}} = \lim_{r\to 0} \frac{r^2\cos\theta\sin\theta}{|r|} = \lim_{r\to 0} |r|\cos\theta\sin\theta = 0$. since $|\cos\theta\sin\theta| \le 1$.

or $0 \le \left|\frac{xy}{\sqrt{x^2+y^2}}\right| \le |y| \to 0$.

(2) $\lim_{(x,y)\to(0,0)} \frac{xy^3 + 2x^2y^4}{x^2+y^6} = \lim_{\substack{(x,y)\to(0,0) \\ x = ky^3}} \frac{ky^6 + 2k^2y^7}{k^2y^6 + y^6} = \lim_{y\to 0} \frac{k + 2k^2y}{k^2+1} = \frac{k}{k^2+1}$

$\Rightarrow$ the limit does not exist.

装订线内不答题！

7. (1) $f(x)=\sum_{n=1}^{\infty}\frac{n+2}{n(n+1)}x^n$

$\left|\frac{u_{n+1}}{u_n}\right|=\left|\frac{(n+3)\,n(n+1)}{(n+2)(n+1)(n+2)}x\right|\to|x|<1 \Rightarrow -1<x<1$

when $x=1$, $\sum_{n=1}^{\infty}\frac{n+2}{n(n+1)}$ diverges since $\frac{\frac{n+2}{n(n+1)}}{\frac{1}{n}}\to 1$, $\sum\frac{1}{n}$ diverges.

when $x=-1$, $\sum_{n=1}^{\infty}(-1)^n\frac{n+2}{n(n+1)}$ converges conditionally. since $\sum\frac{n+2}{n(n+1)}$ diverges

$a_n=\frac{n+2}{n(n+1)}>0$, $\to 0$, $\downarrow$.

$\Rightarrow [-1,1)$

(2) $f(x)=\sum_{n=1}^{\infty}\frac{n+2}{n(n+1)}x^n=\sum_{n=1}^{\infty}\frac{x^n}{n+1}+\sum_{n=1}^{\infty}\frac{2}{n(n+1)}x^n=\sum_{n=1}^{\infty}\frac{x^n}{n+1}+2\sum_{n=1}^{\infty}\frac{x^n}{n}-2\sum_{n=1}^{\infty}\frac{x^n}{n+1}=2\sum_{n=1}^{\infty}\frac{x^n}{n}-\sum_{n=1}^{\infty}\frac{x^n}{n+1}$

$g(x)=\sum_{n=1}^{\infty}\frac{x^n}{n+1}=\frac{1}{x}\sum_{n=1}^{\infty}\frac{x^{n+1}}{n+1}$, $\left(\sum_{n=1}^{\infty}\frac{x^{n+1}}{n+1}\right)'=\sum_{n=1}^{\infty}x^n=\frac{x}{1-x}$

if $x\neq 0$

$\Rightarrow \sum_{n=1}^{\infty}\frac{x^{n+1}}{n+1}=\int\frac{x}{1-x}dx=\int\frac{-1+(1-x)}{1-x}dx=\int\frac{dx}{1-x}-x$

$=-\ln(1-x)-x+C$

let $x=0$. $\Rightarrow C=0$

$\Rightarrow g(x)=-\frac{\ln(1-x)}{x}-1$, if $x\neq 0$.

$h(x)=\sum_{n=1}^{\infty}\frac{x^n}{n}$, $h'(x)=\sum_{n=1}^{\infty}x^{n-1}=\frac{1}{1-x}\Rightarrow h(x)=-\ln(1-x)$

$\Rightarrow f(x)=\begin{cases}-2\ln(1-x)+\frac{\ln(1-x)}{x}+1=1+\left(\frac{1}{x}-2\right)\ln(1-x), & x\neq 0.\\ 0, & x=0.\end{cases}$

8. $\sum_{n=2}^{\infty}\frac{(-1)^{n+1}}{n^p(\ln n)^2}$, $p>0$. let $u_n=\frac{1}{n^p(\ln n)^2}$

converges absolutely if $p>1$

converges conditionally if $0<p\leq 1$.

(1) $\lim_{n\to\infty}\frac{\frac{1}{n^p(\ln n)^2}}{\frac{1}{n^p}}=\lim_{n\to\infty}\frac{1}{\ln n}=0$ if $p>1$ $\Rightarrow\sum_{n=2}^{\infty}\frac{1}{n^p(\ln n)^2}$ converges $\Rightarrow\sum u_n$ converges absolutely.

(2) if $p=1$. $\sum_{n=2}^{\infty}\frac{1}{n(\ln n)^2}$, $f(x)=\frac{1}{x(\ln x)^2}$, $\int_2^{\infty}\frac{dx}{x(\ln x)^2}=-\frac{1}{\ln x}\Big]_2^{\infty}=\frac{1}{\ln 2}$ converges $\Rightarrow\sum u_n$ converges absolutely.

(3) if $0<p<1$. $\lim_{n\to\infty}\frac{\frac{n^p(\ln n)^2}{1}}{\frac{1}{n}}=\lim_{n\to\infty}\frac{n^{1-p}}{(\ln n)^2}=\lim_{n\to\infty}\frac{8n^{2-p}}{2\ln n}=\lim_{n\to\infty}\frac{n^{2-p}}{2}=\infty$, $\sum\frac{1}{n}$ diverges $\Rightarrow\sum\frac{1}{n^p(\ln n)^2}$ diverges $\Rightarrow\sum u_n$ converges conditionally.

装订线内不答题

9. $\lim_{n\to\infty}\left((n^2-n)e^{\frac{1}{n}}-\sqrt{n^4+1}\right)$

let $f(x)=(x^2-x)e^{\frac{1}{x}}-\sqrt{x^4+1}$

$=(x^2-x)\left(1+\frac{1}{x}+\frac{1}{2!x^2}+\cdots+\frac{1}{k!x^k}+\cdots\right)$

$=x^2-\frac{1}{2}-\frac{1}{3x}+\cdots$

$\sqrt{x^4+1}=x^2\sqrt{1+\frac{1}{x^4}}=x^2\left(1+\frac{1}{2x^4}-\frac{1}{8x^8}+\cdots\right)$

$=x^2+\frac{1}{2x^2}-\frac{1}{8x^6}+\cdots$

$\Rightarrow f(x)=-\frac{1}{2}-\frac{1}{3x}+\cdots$

$\Rightarrow \lim_{x\to\infty} f(x)=-\frac{1}{2}$.